Types of Java Classes

Types of Classes

- We've been using classes for a while now
- Almost always with

public class Foo{...}

- Call this an "outer class"

Types of Classes

- Class User
- Take comparator
- An interface
  - We could go write
  - UserComparator
- Just to compare out users “un-naturally”

Types of Classes

- This works
- Kind of clunky though
- Now anywhere I can create a UserComparator without Users

- Lets scope this better
  - Promote encapsulation
  - Reduce Boilerplate

Inner Classes

- Define a class only where we need it
- Inner Class
  - Named Inner class specifically
- class UserComparator{...}
- Fixes out scoping problem
- Just as clunky
  - It's only purpose is to fill an interface

Anonymous Inner Classes

- Let's instantiate the interface directly
- Remember → things after the *new* must have concrete definitions
- So let's write them in-line
- Use curly braces

- Works with all interfaces
  - Even your own

Anonymous Inner Class

- Anonymous Inner-Class
  - Our interface definitions
  - Compiles to a class that implements it

- But... We never actually gave that class a name
- Have we really fixed our problem?

Lambda

- Java has specially syntax for anonymous inner class
- If and only if:
  - The interface has a SINGLE method it
- Lambda expression

Lambda Expression

- Special interface syntax
- (params) - > { body }
- Only works for single method interface

- Most concise way of implementing this
- Don't fill in data types in params
  - Java should already know this